

Preeti De Maurya

Computer Science PhD Candidate | Human-Computer Interaction | UX Research

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Research Profile

Human-centered technology researcher working at the intersection of participatory design, collaborative technologies, and artificial intelligence. My research examines how emerging technologies can support inclusive collaboration and decision-making while preserving human agency. My experience combines qualitative and participatory research with software development, AI/ML, and technology prototyping. Building on this foundation, my future research interests focus on AI-mediated participatory design, User experience engineering and the design of inclusive and ethically grounded collaborative systems.

Research Interests

UX Engineering • Inclusive & Ethical AI • Participatory Design • Organizational Technology & Resilience

Education

PhD, Computer Science <i>New Mexico State University, Las Cruces, NM</i> <i>Focus: HCI, AI/ML, Data Science, Computer Vision</i>	2020–Present
M.S., Automotive Software Engineering <i>Technische Universität Chemnitz, Germany</i> <i>Focus: Software Engineering, Data Science, IoT, Distributed Computing</i>	2017–2019
B.Sc., Computer Science <i>Mulund College of Commerce, Mumbai, India</i> <i>Focus: Data Structures & Algorithms, Software/Web Development, DBMS, Networking</i>	2010–2013

Research Experience

Research Assistant, Participatory Live Experiences Lab (PLEX LAB), New Mexico State University
12/2022–Present

- Conduct participatory design research with community and organizational stakeholders to understand lived experiences, needs, and opportunities for technology design.
- Lead and facilitate participatory design workshops with teachers, disaster-management experts, and other domain stakeholders.
- Conduct qualitative research on mixed-reality applications at a Hispanic-Serving Institution, examining experiences of students from diverse cultural backgrounds.
- Implement research infrastructure using Docker and Kubernetes to support scalable deployment.

Graduate Assistant Researcher, FinTech Lab, New Mexico State University **07/2022–11/2023**

- Developed a real-estate price forecasting application combining machine learning and full-stack software development.
- Applied ARIMA-based machine learning models in Python and developed the application interface with ReactJS, Django backend services, and MySQL.

Research Assistant, New Mexico State University **08/2021–07/2022**

- Independently designed and developed Hack the Land Cover, a NASA EPSCoR-funded collaborative platform for collecting volunteer-provided image labels.
- Developed the web interface using Vue.js and Canvas UI and stored labeled data using MongoDB.
- Applied image segmentation and image-recognition machine-learning approaches to identify land-cover types.

Research Internship / Thesis, IAV GmbH, Germany **07/2018–12/2019**

- Applied machine-learning clustering methods to identify correlations among nonlinear parameters associated with maximum emissions in real-time combustion-engine testing.

Professional Experience

Lead Software Engineer, Infinite Devices GmbH, Germany **02/2020–07/2021**

- Developed backend and IoT services for Infinimesh, including MQTTv5 support for proximity-sensor communication.
- Worked with Golang, Redis, Kafka, MQTT, Dgraph, gRPC, Wireshark, Docker, Kubernetes, GitHub Webhooks, and CircleCI.
- Developed CI/CD pipelines and supported distributed application deployment.

Software Engineer, IAV GmbH, Germany **07/2019–06/2020**

- Implemented Time Sensitive Networking diagnostics and network protocols for Ethernet and CAN environments.
- Developed diagnostic test scripts using CANoe and C#.

Software Developer, Syntel Ltd., India **09/2013–01/2016**

- Worked as a full-stack developer and software development/test engineer for global clients including AMEX and FedEx.
- Developed software using Java, Groovy, JavaScript, Oracle, and MySQL.

Student Mentoring

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- Alejandro Rodriquez, 2025, *3D Tangibles for Lo-Fi Prototyping*
 - Emma Jackson, 2023, *Study on Designing Games for Students with Dyslexia*

Selected Publications

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- Research Paper submitted to GROUP 2027 — Accepted (Survey of Ethical Practices for Participatory Design Involving Vulnerable Populations).
 - De Maurya, P., Ul Haque Choyon, R., Platt, T., Hamilton, B., Contreras Aguirre, H. C., Carrillo, L. R. G. “Fostering an Inclusive Community Among Electrical and Computer Engineering Students with Mixed-Reality Technologies at a Hispanic-Serving Institution.” *ASEE CoNECD 2025*. Accepted.
 - Jafari, O., Maurya, P., Islam, K. M., & Nagarkar, P. “Optimizing Fair Approximate Nearest Neighbor Searches using Threaded B+-Trees.” *14th International Conference on Similarity Search and Applications (SISAP)*, 2021.
 - Ibrahim, N., Maurya, P., Jafari, O., & Nagarkar, P. “A Survey of Performance Optimization in Neural Network-Based Video Analytics Systems.” *arXiv*, 2021.
 - Jafari, O., Maurya, P., Nagarkar, P., Islam, K. M., & Crushev, C. “A Survey on Locality Sensitive Hashing Algorithms and their Applications.” *arXiv*, 2021.
 - Maurya, P., Jafari, O., Thatte, B. et al. “Building a comprehensive NER model for Satellite Domain.” *SN Computer Science* 3, 199 (2022).

Grants

Collaborative Research: HCC: Medium: Connecting Practitioners to Design: Methods and Tools for Live Participatory Design Fiction, NSF Award Number 2425383

Role: Student Researcher

HCC: Medium: Organizational Resilience Making with Technology, NSF Award Number 2403345

Role: Student Researcher

Technical Skills

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- **Programming:** Java, Python, Go, Groovy, JavaScript, Vue.js, ReactJS, Node.js
 - **AI/ML:** ARIMA, clustering, image segmentation, image recognition, computer vision
 - **Data & Backend:** MySQL, Oracle, PostgreSQL, Redis, MongoDB/NoSQL, Dgraph/GraphQL, Elasticsearch, Django, Flask

- **Distributed & Cloud:** Docker, Kubernetes, Kafka, gRPC, REST/SOAP, MQTT, CI/CD, CircleCI, Git
- **Research Methods:** Participatory Design, qualitative studies, interviews, focus groups, collaborative prototyping, design thinking
- **Qualitative UX & HCI:** User interviews, usability testing, participatory design, co-design, qualitative thematic analysis, workshop facilitation, UX research synthesis

Workshops and User Studies Organized

- De Maurya, P., Ul Haq Choyon, R., Platt, T., Carrillo, L. R. G., Contreras Aguirre, H. C., Taylor, M., & Hamilton, B. (2024). "Designing Mixed Reality for Social Learning in Wisdom Communities at Hispanic Serving Institutions."
- De Maurya, Jackson Emma (2024), "Study on Designing Games for Students with Dyslexia."
- De Maurya, Cole Jackson, Hamilton Bill, Lalone Nick (2025), "Participatory Design with Emergency Management Experts to Design Emergency Management System".
- De Maurya, Rodrigues Alejandro, Hamilton Bill, "Participatory design and prototyping of 3D tangibles for undergrad students to improve their experiences of summer programs".

Workshops Attended

- Nebeling Michael, Oki Mika, Gelsomini Mirko, Hayes Gillian R, Designing Inclusive Future Augmented Realities, CHI 2024.
- Houben Maarten, Lee Minha, Foley Sarah, HCI Research in Sensitive Settings: Learning Researcher Reflexivity, Ethical Conduct and Empathy in Participatory Design Approaches, CHI 2024.
- Jeremy A. Magruder Waisome, Dr. Jerrod A Henderson, Elevating Community Visions of ECSJ through Photovoice, ASEE 2025.